College of Information Science

CYBR 312 – Python Scripting in Cybersecurity
The University of Arizona
College of Information Science
Fall 2026 (7W1)

Instructor:	Cameron Covington
Email:	ccovington@arizona.edu
Office:	Remote
Office Hours:	By appointment
Course Delivery Method:	Online
Location & Time:	Tuesdays, 7PM CT

COURSE DESCRIPTION:
CYBR 312 will provide students with an introduction to the practical applications of Python programming in support of cybersecurity and digital investigation activities. The course will provide students with a solid foundation of the use of python language elements along with a practical understand on how to develop, debug, execute and deploy python scripts.

Prerequisites: *None*

REQUIRED TEXTS:

- Title: Automate the Boring Stuff with Python (2nd Edition)
  - Author: Al Sweigart
  - ISBN: 978-1593279929
  - Publisher: No Starch Press
  - https://www.amazon.com/Automate-Boring-Stuff-Python-2nd/dp/1593279922
  - Also available for student download: https://ebookcentral.proquest.com/lib/UAZ/detail.action?docID=6001752

STUDENT LEARNING OUTCOMES:
Upon completion of this course students will be able to:

- Define and demonstrate the basic building blocks of the Python scripting

language.
- Describe and demonstrate how to use and experiment with a Python integrated development environment.
- Describe and demonstrate access to and inventory of a file system.
- Demonstrate the use of key Python built in data types including strings, lists, sets, and dictionaries to cybersecurity-related challenges.
- Develop and demonstrate scripts that can search, parse, and index text and binary files to uncover key evidence.
- Describe and demonstrate how to access and utilize key Python standard libraries, including but not limited to OS, SYS, CSV, JSON, RE, PATHLIB, HASHLIB, LOGGING, ARGPARSE, and PICKLE.
- Describe and demonstrate how to discover, install, and utilize 3rd party Python libraries.

COURSE OBJECTIVES

During this course students will:

- Use hands-on coding exercises to demonstrate they have mastered the application of security scripting.
- Experiment, debug, enhance and describe the operation and applicability of security scripts for Cyber Operations.
- Develop Python scripts targeting specific cybersecurity challenges.

EXPECTED COURSE WORKLOAD

Students should expect a minimum of 4 hours of lecture and an additional 14 hours of homework/exercises/reading each week for this course. Live synchronous lectures will be held each week. If a student cannot attend the lecture, the lectures are recorded and can be viewed by the student asynchronously.

REQUIREMENTS:

Requirements for the course are 7 Quizzes, 7 Scripting Exercises and a Final Scripting Project. Students will be expected to draw on course lectures and assigned readings to complete all course assignments and Exercises. *No late work will be accepted so please ensure you allot enough time to complete and turn in your work on time.*

GRADING:

The final grade in the course will be based upon:

Quizzes – 350 Points	**35%**	**See Course Schedule/D2L**
(7 Quizzes - 50 Points Each)		
Programming Exercises – 350 Points	**35%**	**See Course Schedule/D2L**
(7 Scripting Assignments – 50 Points Each)		
Final Scripting Exercise – 300 Points:	**30%**	**See Course Schedule/D2L**
(Final Examination)		

GRADING SCALE			DISTRIBUTION	
A	90 to 100 %		900 – 1000 Points	A
B	80 to 89%		800 – 899 Points	B
C	70 to 79%		700 – 799 Points	C
D	60 to 69%		600 - 699 Points	D
E	Below 60%		0 – 599 Points	E

COURSE SCHEDULE:

Week 1 – Introduction to Python and the Integrated Development Environment (8/24-8/30)

- **Week 1 Learning Objectives** – Upon completion students will be able to describe and demonstrate:
  - Leverage the Virtual Learning Environment and Desktop that is provided to:
  - Experiment with provided sample cyber security scripts.
  - Experiment with:
    - Standard data types: Boolean types, strings, integers applied to relevant cyber security data.
  - Examine the contents and type of variables/objects using the Integrated Development Environment (IDE).
  - Apply built-in functions such as hex, bin, int, and format to examine cybersecurity related data in hexadecimal, binary and other key formats.

- **Week 1 Assignments**
  - Textbook: Chapter 1: Read, Study and experiment with each of examples provided in the chapter.
  - Watch the Week One Videos
  - Attend the WK1 Online Lecture and Demonstration
  - Complete Quiz 1 Python Setup and Interaction
  - Scripting assignment 1: Demonstrate the use of the Python shell within WingIDE to solve simple challenges.

Week 2 – Cybersecurity Script Structure (8/31-9/6)

- **Week 2 Learning Objectives** – Upon completion students will be able to describe and demonstrate:
    - The use of decision-making structures in python
        - if, elif, else
        - The proper use of indentation
        - The application of iteration and loops applied to cyber security objects.
        - Creation of string objects along and the examination of those objects to extract probative cyber security and forensic evidence that is usable by cyber investigators.

- **Week 2 Assignments**
    - Textbook: Chapter 2 and 9: Read, Study and experiment with the examples provided.
    - Watch the Week Two Videos
    - Attend the WK2 Online Lecture and Demonstration
    - Complete Quiz 2 – Decision Making and String Processing
    - Scripting assignment 2 – Creating your first script.

Week 3 – Accessing the local file system (9/7-9/13)

- **Week 3 Learning Objectives** – Upon completion students will be able to describe and demonstrate:

  Develop solid understanding of filesystem structures examined from a cybersecurity point of view.

    - Differentiate relative, absolute file paths, archive, hidden and system files as a pre-cursor to cyber-investigations.
    - Search and inventory files and file system attributes within a given directory to obtain a Cyber Operations baseline.
    - Develop a foundational understanding of cryptographic hashing and the myriad of hashing algorithms available. Learn how cryptographic one-way hashes are applied to cybersecurity, forensics, and incident response situations.
    - Develop a cryptographic hashing script that processes files in support of Cyber Operations and indication of compromise.

- **Week 3 Assignments**
    - Textbook: Chapter 10, read, study and experiment. Then attempt the challenge problem posed on page 248 that identifies large files.
    - Watch the Week Three Videos
    - Attend the WK3 Online Lecture and Demonstration
    - Complete Quiz 3 Python filesystem processing
    - Scripting assignment 3 – Create a filesystem inventory.

Week 4 – Searching / Indexing text and binary files (9/14-9/20)

- **Week 4 Learning Objectives** – Upon completion students will be able to describe and demonstrate:
  - Filesystem searching from a Cyber Operations perspective.
  - Understand the limitations of keyword searches within Cyber Operations/forensics and how to improve their effectiveness during investigation and incident response.
  - Learn how indexing can improve search effectiveness and provide insights during investigation and cyber ops.
  - Learn how to process large files and extract probative content in support of cyber investigations.
  - Create a cyber security script that can identify files based on extension, content and a targeted list of cryptographic one-way hashes typically used in the identification of compromise and/or concealment of contraband.

- **Week 4 Assignments**
  - Textbook: Chapter 4, Read, Study and Experiment with examples provided.
  - Read materials and watch videos provided by the professor.
  - Attend the WK4 Online Lecture and Demonstration
  - Complete Quiz 4 Python Search and Index
  - Scripting Assignment 4 – Searching for files that provide probative value during Cyber Operations and investigations.

Week 5 – Using Rainbow Tables during Cyber Operations (9/21-9/27)

- **Week 5 Learning Objectives** – Upon completion students will be able to describe and demonstrate:
  - What Cyber Operations rainbow tables are, and the threat that they pose to password discovery.
  - How modern passwords are protected using salts to mitigate static rainbow table cyber-attacks.
  - How to create dynamic rainbow tables to attack salted passwords when suspect tendencies are known.
  - How to apply key libraries to generate combinations, permutations, and products to generate brute force passwords for use in password cracking Cyber Ops.

- **Week 5 Assignments**
  - Textbook: Chapter 5 Read, Study and experiment with examples provided.
  - Read materials and watch videos provided by the professor.
  - Attend the WK5 Online Lecture and Demonstration
  - Complete Quiz 5 – Password Cracking
  - Scripting Assignment 5 – Utilize a baseline dynamic rainbow table script to discover and (crack) the passwords.

Week 6 – Script logging within Cyber Operations (9/28-10/4)

- **Week 6 Learning Objectives** – Upon completion students will be able to

describe and demonstrate:

- The importance of logging during Cyber Operations activity, forensic investigations, and incident response.
- The use of logging during Cyber Operations
    - Apply systematic logging of security script activity that reports, information, warning, and critical log events.
    - Build a standard reusable model to post log events that includes the event level, universal coordinated time (UTC) events, along with a descriptive message relating to the event.

- **Week 6 Assignments**
    - Textbook: Chapter 11 Pages: 255-260 **Logging section**. Read, Study and Experiment with examples.
    - Read materials and watch videos provided by the professor.
    - Attend the WK6 Online Lecture and Demonstration
    - Complete Quiz 6 – Logging
    - Scripting assignment 6 – Utilize the cyber security script provided by the professor and add the ability to generate a detailed log of the script operation.

Week 7 – Image Analysis in Cyber Ops and Forensics (10/5-10/11)

- **Week 7 Learning Objectives** – Upon completion students will be able to describe and demonstrate:

    - Identify the importance of photographic image data within Cyber Operations.
    - Search the file system for photographic images based on file content NOT file extension or name.
    - Extract location, image format, dimensions, and image type of all discovered images to be used by Cyber Operations analysts
    - Identify, alert and report malicious mismatches between filename/extension and the discovered images as a warning to cyber investigators.
    - Report image search results

- **Week 7 Assignments:**
    - Textbook: Chapter 16 and 19 Read, Study and Experiment with the examples provided.
        - Read materials and watch videos provided by the professor.
        - Quiz 7 – Searching and Interrogating Image Files
        - Scripting Assignment 7 – Creating an image search security script for use in cyber operation investigations.

Final Scripting Exercise (due 10/14)

- A final scripting assignment will be provided.

University of Arizona Final Examination Policy: Final Exams | Office of the Registrar (arizona.edu) and Final Exam Schedule http://www.registrar.arizona.edu/schedules/finals.htm

Grading Policy:
University policy regarding grades and grading systems is available at:
http://catalog.arizona.edu/policy/grades-and-grading-system

Requests for incompletes (I) and withdrawal (W) must be made in accordance with university policies which are available at http://catalog.arizona.edu/policy/grades-and-grading-system#incomplete and http://catalog.arizona.edu/policy/grades-and-grading-system#Withdrawal respectively.

Classroom Behavior Policy:
To foster a positive learning environment, students and instructors have a shared responsibility. We want a safe, welcoming, and inclusive environment where all of us feel comfortable with each other and where we can challenge ourselves to succeed. To that end, our focus is on the tasks at hand and not on extraneous activities (e.g., texting, chatting, reading a newspaper, making phone calls, web surfing, etc.).

Threatening Behavior Policy:
The UA Threatening Behavior by Students Policy prohibits threats of physical harm to any member of the University community, including to oneself. See: http://policy.arizona.edu/education-and-student-affairs/threatening-behavior-students

Accessibility and Accommodations:
At the University of Arizona we strive to make learning experiences as accessible as possible. If you anticipate or experience barriers based on disability or pregnancy, please contact the Disability Resource Center (520-621-3268, https://drc.arizona.edu/) to establish reasonable accommodations.

Code of Academic Integrity as described in the UA General Catalog:
Students are encouraged to share intellectual views and discuss freely the principles and applications of course materials. However, graded work/exercises must be the product of independent effort unless otherwise instructed. Students are expected to adhere to the UA Code of Academic Integrity as described in the UA General Catalog. See: https://deanofstudents.arizona.edu/policies/code-academic-integrity

The University Libraries have some excellent tips for avoiding plagiarism available at:
http://new.library.arizona.edu/research/citing/plagiarism

All written work is submitted to *Turnitin*, an anti-plagiarism application that compares submitted assignments to a comprehensive database that includes all published and internet sources as well of hundreds of thousands of student papers. You will be able to see for yourself what percentage of your written work is considered "non-original" once you have submitted it to the appropriate Dropbox in D2L. My position on plagiarism is non-negotiable—any written work that is plagiarized will result in a failure for that assignment and possibly the course. Please familiarize yourself with the rules regarding plagiarism.

Selling class notes and/or other course materials to other students or to a third party for resale is not permitted without the instructor's express written consent. Violations to this and other course rules are subject to the Code of Academic Integrity and may result in course sanctions. Additionally, students who use D2L or UA e-mail to sell or buy these copyrighted materials are subject to Code of Conduct Violations for misuse of student e-mail addresses. This conduct may also constitute copyright infringement.

UA Nondiscrimination and Anti-Harassment Policy:
The University is committed to creating and maintaining an environment free of discrimination; see http://policy.arizona.edu/human-resources/nondiscrimination-and-anti-harassment-policy. Our classroom is a place where everyone is encouraged to express well-formed opinions and their reasons for those opinions. We also want to create a tolerant and open environment where such opinions can be expressed without resorting to bullying or discrimination of others.

Absence and Class Participation Policy:
The UA's policy concerning Class Attendance, Participation, and Administrative Drops is available at: http://catalog.arizona.edu/policy/class-attendance-participation-and-administrative-drop
The UA policy regarding absences for any sincerely held religious belief, observance or practice will be accommodated where reasonable, http://policy.arizona.edu/human-resources/religious-accommodation-policy.

Absences pre-approved by the UA Dean of Students (or Dean Designee) will be honored. See: https://deanofstudents.arizona.edu/absences

Additional Resources for Students:

UA Academic policies and procedures are available at:
http://catalog.arizona.edu/policies

Student Assistance and Advocacy information is available at:
Student Assistance | Dean of Students Office (arizona.edu)

Academic Advising:
If you have questions about your academic progress this semester, please reach out to your academic advisor (https://advising.arizona.edu/advisors/major). Contact the Advising Resource Center (https://advising.arizona.edu/) for all general advising questions and referral assistance. Call 520-626-8667 or email to advising@.arizona.edu

Life Challenges:
If you are experiencing unexpected barriers to your success in your courses, please note the Dean of Students Office is a central support resource for all students and may be helpful. The Dean of Students Office can be reached at (520) 621-2057 or DOS-deanofstudents@email.arizona.edu.

Physical and Mental-Health Challenges:
If you are facing physical or mental health challenges this semester, please note that Campus Health provides quality medical and mental health care. For medical appointments, call (520) 621-9202. For After Hours care, call (520) 570-7898. For the Counseling & Psych Services (CAPS) 24/7 hotline, call (520) 621-3334.

Confidentiality of Student Records
http://www.registrar.arizona.edu/personal-information/family-educational-rights-and-privacy-act-1974-ferpa?topic=ferpa

***Subject to Change Statement**
Information contained in the course syllabus, other than the grade and absence policy, may be subject to change with advance notice, as deemed appropriate by the instructor.